

LYNCHBURG, VA, USA

WMO: 724100

Lat: 37.321N

Lon: 79.207W

Elev: 287

StdP: 97.93

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13733

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.6	-7.3	-19.6	0.7	-5.6	-17.0	0.9	-3.8	8.7	8.6	7.9	7.5	1.6	300	0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	33.2	23.1	31.9	22.7	30.7	22.2	24.6	30.4	24.0	29.4	23.4	28.5	2.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.3	26.7	22.4	17.7	26.1	21.9	17.1	25.6	75.8	30.6	73.2	29.6	70.9	28.5	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.6	6.6	5.6	DB	-14.9	35.0	3.4	1.7	-17.3	36.2	-19.3	37.2	-21.2	38.2	-23.6	39.4	(4)
			WB	-15.2	26.1	2.8	0.8	-17.2	26.7	-18.9	27.2	-20.4	27.7	-22.5	28.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.6	2.2	3.9	8.2	13.8	18.0	22.4	24.3	23.7	20.1	14.0	8.2	4.1
	DBStd	8.91	5.64	5.20	5.31	4.63	4.02	2.96	2.46	2.46	3.45	4.40	4.76	4.88
	HDD10.0	842	251	179	99	16	1	0	0	0	0	13	91	193
	HDD18.3	2342	501	403	315	148	57	5	1	1	22	145	305	441
	CDD10.0	2169	8	9	44	130	250	372	444	425	303	138	36	11
	CDD18.3	629	0	0	1	13	48	127	187	167	74	12	1	0
	CDH23.3	5479	1	1	22	175	445	1102	1640	1399	575	112	7	1
	CDH26.7	1946	0	0	2	38	116	381	672	531	183	24	0	0
Wind	WSAvg	2.2	2.5	2.5	2.8	2.8	2.3	1.9	1.7	1.6	1.8	1.9	2.1	2.1
Precipitation	PrecAvg	1036	76	78	93	80	98	85	107	86	87	80	76	82
	PrecMax	1516	202	154	235	202	230	253	262	286	306	290	223	182
	PrecMin	674	12	14	19	7	22	12	29	24	0	0	8	8
	PrecStd	222	43	38	45	39	51	58	53	52	69	65	45	41
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.8	21.5	25.9	30.1	31.3	33.8	35.4	34.5	32.7	29.3	24.2	20.9
		MCWB	14.6	14.2	15.8	18.0	21.4	23.3	23.8	23.7	21.5	20.8	16.3	14.1
	2%	DB	16.7	18.0	23.0	27.6	29.5	32.0	33.5	32.8	30.7	26.3	21.0	17.4
		MCWB	13.3	11.6	13.9	16.9	20.6	22.7	23.4	23.1	21.5	18.8	14.4	13.7
	5%	DB	14.2	15.4	20.2	25.4	27.9	30.4	31.9	31.3	28.8	23.9	18.9	14.7
		MCWB	10.8	10.1	12.5	16.0	19.8	22.3	23.1	22.7	20.7	17.5	13.8	11.2
	10%	DB	11.3	12.7	17.6	22.8	26.0	29.0	30.4	29.7	26.9	21.8	16.6	12.4
		MCWB	7.8	7.9	11.4	14.8	19.0	21.6	22.7	22.1	20.0	17.0	12.4	8.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.6	16.2	17.1	20.1	22.9	24.6	25.6	25.6	24.0	22.6	18.4	17.6
		MCDB	18.4	19.4	23.6	25.7	28.5	30.6	32.0	31.6	29.0	25.6	20.2	19.8
	2%	WB	14.2	13.6	15.7	18.6	21.9	23.8	24.8	24.5	23.0	21.0	16.8	14.7
		MCDB	16.3	16.3	20.7	24.3	27.4	29.7	30.6	30.1	27.4	24.1	19.5	16.4
	5%	WB	11.2	11.1	14.2	17.4	20.9	23.1	24.1	23.9	22.1	19.5	15.0	12.0
		MCDB	13.2	14.1	18.5	22.9	26.1	28.7	29.5	28.9	26.2	22.7	17.8	13.8
	10%	WB	7.9	8.3	12.4	16.1	20.0	22.5	23.5	23.1	21.3	17.6	13.0	9.2
		MCDB	10.9	11.9	16.6	21.0	24.5	27.6	28.6	27.7	24.8	20.5	15.8	11.6
Mean Daily Temperature Range	5% DB	MDBR	10.1	10.8	11.9	12.9	12.0	11.4	11.0	11.0	11.1	12.0	11.9	10.1
		MCDBR	13.5	15.3	15.7	16.3	14.2	13.2	13.1	13.1	13.8	14.4	15.0	13.3
		MCWBR	9.9	10.3	8.9	7.6	6.6	5.6	4.6	5.0	5.7	7.4	9.2	9.5
	5% WB	MCDBR	12.0	13.3	13.5	13.4	12.1	11.6	11.4	11.1	11.0	11.2	11.7	11.6
		MCWBR	10.0	10.1	8.5	7.3	6.0	5.2	4.6	4.8	5.1	6.6	8.9	9.5
Clear-Sky Solar Irradiance	taub	0.299	0.315	0.348	0.388	0.430	0.476	0.506	0.506	0.399	0.357	0.318	0.302	
	taud	2.510	2.454	2.408	2.314	2.211	2.125	2.064	2.040	2.337	2.443	2.515	2.537	
	Ebn at Noon	895	920	916	893	856	814	785	775	850	859	864	868	
	Edh at Noon	79	94	109	127	143	156	164	165	116	95	78	72	
All-Sky Solar Radiation	RadAvg	2.35	3.09	4.13	5.13	5.69	6.20	5.85	5.30	4.39	3.53	2.63	2.04	
	RadStd	0.17	0.37	0.27	0.39	0.49	0.42	0.37	0.26	0.44	0.41	0.20	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+126	+82
(47)	Regional (46 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140	+78

Nomenclature: See separate page